

AI & AUTOMATION · GTM SYSTEMS ENGINEERING

Patrick Diamitani

GTM AI & Automation Manager · AI & Automation lead at Atlas HXM.

I build production AI automations that ship — and the systems that let a 50+ person GTM org run on them.

Targeting: Director of AI & Automation · Applied-AI / Forward-Deployed Engineering · Founding GTM-Engineering roles

pdiamitani@atlashxm.comlinkedin.com/in/diamitani

Chicago, IL

IMPACT AT A GLANCE

8+

production automations live across the GTM stack

~242

pipeline deals auto-screened for EOR compliance, every one auditable

0→stable

post-sales workflow recovered from 21/25 failing to reliable end-to-end

515K

credit spend spike traced to root cause & governed with a reusable template

<30 min

idea → fully-scoped project, down from 4–8 hours

30+

reusable Claude skills & agents shipped for the whole GTM team

TECHNOLOGY & PLATFORMS

ORCHESTRATION

n8n

automation platform that chains apps so steps fire automatically — no human clicking buttons

Webhooks & MCP

AI / LLM

Azure OpenAI (GPT-4o / GPT-5-chat)

the reasoning layer that reads messy text and classifies or drafts

Claude + Claude Skills / Agent SDK

the platform used to build the reusable skill & agent library

event triggers and tool-connectors that let systems call each other in real time

Custom JavaScript engines

deterministic rules written in code, so decisions are repeatable and auditable

GTM DATA & CRM

HubSpot API

the CRM system of record — deals, contacts, properties, workflows

Clay

data-enrichment workspace that builds and enriches prospect lists

Amplemarket & ZoomInfo

contact discovery and firmographic enrichment sources

Avoma

call-recording & transcription intelligence

Factors.ai

first-party website intent & account signals

Retrieval & grounding

feeding models real source data so outputs are specific, not generic

INFRA & DELIVERY

Azure Blob Storage

cloud file store used to serve 250+ country data files deterministically

Microsoft 365 (Outlook, Teams, Excel, OneDrive)

delivery + human-readable mirrors

Onspring

internal governance/compliance data source

PORTFOLIO OF BUILT WORK

FLAGSHIP · LIVE IN PRODUCTION

EOR Compliance Checker (v5)

Auto-screens every sales-pipeline deal against a 30-sector compliance worksheet the moment it enters — and writes an auditable verdict back to the CRM.

n8n · Azure OpenAI GPT-4o · HubSpot API · Avoma API

LIVE · MAJOR RELIABILITY TURNAROUND

Post-Sales Call Automation

After every sales call, reads the transcript and auto-generates a rep-ready follow-up package: consultative email, missed-questions analysis, competitor research, LinkedIn draft, and a logged CRM note.

n8n cloud · Avoma webhook · Azure OpenAI · Azure Blob Storage (250+ country files)

BUILT · REQUESTED BY VP GROWTH

Pre-Sales Call Automation

The moment a meeting is booked, sends the rep a concise pre-call prep note so they walk in already knowing where they're most likely to close.

HubSpot workflows · n8n · Factors.ai Journey API · Azure OpenAI

LIVE · RECOVERED A WHOLE CHANNEL

Chatbot MQL Automation + Attribution

When a website chat ends without a booking, an AI classifies MQL vs. non-MQL, drafts the right follow-up, routes it, and — critically — makes the chatbot a measurable, attributable GTM channel.

HubSpot chatbot + workflows · n8n · Azure OpenAI · HubSpot CRM API

DEPLOY-READY

Partner PQL Workflow

Classifies inbound partner-form submissions, drafts a partner-specific email, and routes a task to the partnerships owner automatically.

n8n · Azure OpenAI (gpt-5-chat) · HubSpot API

BUILD-READY · APPROVAL-GATED

Entity Readiness Change Report

Every Monday, diffs the company's country hire-readiness against last week and emails Growth only the countries that flipped — as a locked two-column table.

n8n · Onspring · n8n Data Table · Microsoft Excel / OneDrive

ONGOING · OWNS THE PIPELINE

SDR-Agency Prospecting Enablement (AGS)

Runs the full prospecting supply chain for an outsourced SDR agency — clean, enriched, CRM-safe lists activated on a weekly cadence.

Clay · Amplemarket · ZoomInfo · HubSpot API

STRATEGY + BUILD

Outbound Signals & Intent Engine

A defined library of fit + intent signals, a 0–100 account scoring model, and composite “plays” that stack signals into prioritized, repeatable outbound motions.

Clay · Factors.ai · HubSpot · Scoring model (0–100)

SHIPPED · COST + GOVERNANCE WIN

Clay Credit Governance & Cost Control

Traced a runaway enrichment-credit spike to its real sources and shipped a reusable template + guardrails that cut per-row cost and stop the leak repeating.

Clay API · Excel data modeling · PDF one-sheeter for leadership

META-SYSTEM · FRAMEWORK

New Project Automation System (Center of Excellence)

Turns any raw input — notes, files, links, a one-line idea — into a complete, leadership-ready project package: knowledge base, PRD, architecture diagram, reporting framework, and a phased build plan.

Claude / Agent SDK · n8n · Notion · Asana

PLATFORM · BUILT FOR 50+ REPS

GTM Skill & Agent Library (30+)

A library of reusable AI skills and agents that put every automation above into the hands of non-technical reps and marketers through plain-English commands.

Claude Skills / Agent SDK · n8n · Clay · HubSpot

EOR Compliance Checker (v5)

Auto-screens every sales-pipeline deal against a 30-sector compliance worksheet the moment it enters — and writes an auditable verdict back to the CRM.

THE PROBLEM

Reps advanced deals to contracting & KYC without knowing if the role or industry was even eligible for EOR. Compliance only caught blockers at contract review — after the prospect was ready to sign. Lost deals, wasted KYC, eroded trust.

WHAT I BUILT

- A 46-node n8n production workflow (real-time webhook + hourly safety-net sweep + on-demand batch re-screen).
- A hybrid engine: the LLM only maps messy CRM data to one of 30 canonical sectors; a deterministic JavaScript matrix makes every tier decision.
- A three-axis compliance framework (viability tier + advisory flags + hard screen-outs) transcribed faithfully from the official worksheet.
- A report-only dry-run harness that tests any future rule change against the entire pipeline with zero writes.

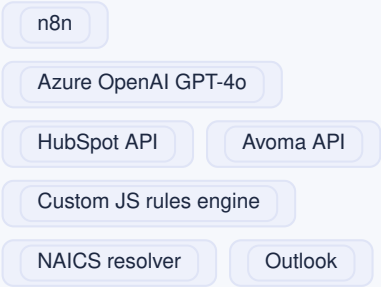
SYSTEMS THINKING

The core design decision — rules live in code, not in the model's head — is what makes it durable. The LLM does only what it's good at (reading unstructured text); every verdict is deterministic, repeatable, and cites its own reasoning. Edge cases were reasoned about explicitly: pass-through detection (a staffing firm placing workers into oil & gas is screened as oil & gas), semantic sector aliasing, and a "needs-title" hold instead of guessing on thin data.

OUTCOME

- Live in production, screening the full Sales Pipeline (~242 open deals).
- Corrected 103 deals wrongly stuck in "review required" while tightening genuine ineligibility detection (Tier 3 caught up from 7 → 15, all traceable to a named worksheet trigger).
- 16 unit tests + full dry-run validation + 238/238 clean production writes before go-live.

STACK



WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
A const-reassignment bug surfaced on the first live production test	Caught immediately by the verification step; fixed and re-tested green. Prevention: the dry-run harness now gates every change.
Five analytics properties silently failed to write (didn't exist in the CRM)	Folded that content into the audit-note field instead. Prevention: validate property existence before writing.

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
Old prompt let thin data demote clean sectors	Re-architected so confidence is a report field that never changes the tier.

Post-Sales Call Automation

After every sales call, reads the transcript and auto-generates a rep-ready follow-up package: consultative email, missed-questions analysis, competitor research, LinkedIn draft, and a logged CRM note.

THE PROBLEM

Reps spent time after every call on follow-up research and writing. The automation that was supposed to do it was failing 21 of 25 recent runs — a dead vector-search index left four AI agents with no country context.

WHAT I BUILT

- An always-on n8n cloud workflow triggered by a call-platform webhook, so it fires regardless of whose laptop is open.
- Four AI agents (follow-up email, missed questions, competitor research, LinkedIn post) grounded in real country data.
- A deterministic replacement for the broken search: extract countries → ISO-3 codes → fetch each file directly from blob storage → inject into every agent.

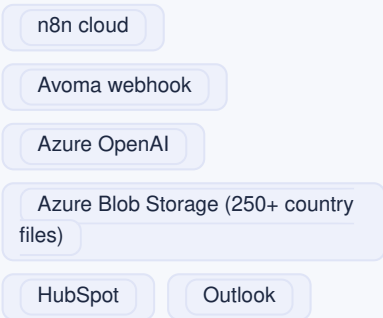
SYSTEMS THINKING

This is a textbook resilience rebuild. Rather than recreate a fragile index (which needed permissions Patrick didn't have), the whole failure point was designed out: country code → exact file, no fuzzy search, no credentials to rotate, and a graceful fallback so the email still sends if a file is missing. Failure modes were mapped end-to-end — future/scheduled meetings, broken item-lineage, context bloat — and each got an explicit guard.

OUTCOME

- Recovered from 0/25 runs reaching the email step to reliable end-to-end delivery.
- Cut context bloat from 20 countries / 882KB to intent-only 6 countries / 295KB per call — cheaper and higher signal.
- Restored specific, grounded output (real severance formulas, social-security rates) over generic filler.

STACK



WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
84% of failures: dead Azure AI Search index (DNS not-found)	Removed the dependency entirely; replaced with deterministic blob fetch. Prevention: prefer deterministic lookups over services that can silently die.
Silent dead-ends on scheduled/future meetings	Added a meeting-completion guard + bounded retry budget (10→20 polls) + explicit exit nodes.

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
Broken paired-item lineage dropped the transcript downstream	Carried the base object forward and converted 28 fragile item-references to a stable form.

03

BUILT · REQUESTED BY VP GROWTH

Pre-Sales Call Automation

The moment a meeting is booked, sends the rep a concise pre-call prep note so they walk in already knowing where they're most likely to close.

THE PROBLEM

Reps burned time on manual pre-call research; time-zone-based daily checks risked missing calls booked across regions.

WHAT I BUILT

- HubSpot meeting-booked webhook → n8n → enrichment → one-paragraph prep note emailed to the assigned rep.
- Enrichment pulls first-party intent (Factors Journey API), company data, and the most recent contact message.
- A guardrail against duplicate sends; the summary is pre-built instantly and timed to arrive a set number of hours before the call.

SYSTEMS THINKING

Chose an event trigger (meeting booked) over a scheduled daily sweep specifically to defeat the time-zone failure mode — a systems decision made before the first line was built, and documented with an explicit risk/mitigation in the brief.

OUTCOME

- Scoped, brief-approved, and built against a clear definition-of-ready (webhooks fire cleanly, AI uses only factual data — no hallucinations).

STACK

HubSpot workflows

n8n

Factors.ai Journey API

Azure OpenAI

WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
Risk: a daily check could miss calls in other time zones	Mitigation designed in up front — trigger on booking, dedupe, and send relative to call time regardless of zone.

Chatbot MQL Automation + Attribution

When a website chat ends without a booking, an AI classifies MQL vs. non-MQL, drafts the right follow-up, routes it, and — critically — makes the chatbot a measurable, attributable GTM channel.

THE PROBLEM

The chatbot (an AI customer agent) was capturing conversations, but the attribution flag was never being stamped — so 364 chatbot contacts going back years were invisible to every report and list. The channel looked dead when it wasn't.

WHAT I BUILT

- An n8n workflow that classifies intent (Azure OpenAI), generates a tailored follow-up, and routes MQL vs. nurture — with an idempotency flag so nothing double-processes.
- Three coordinated HubSpot workflows: chat-end trigger, MQL owner-assignment + internal notification, and non-MQL system email with resources.
- An attribution-stamp workflow + a 364-contact API backfill (364/364 updated, 0 errors) and five smart lists to make the channel reportable.

SYSTEMS THINKING

Diagnosed the root attribution failure before building anything (a data audit found the capture-vs-stamp gap), then designed for reliability: UUID session IDs, an idempotency checkpoint, and system-inbox sending so delivery never depends on a person.

OUTCOME

- 364 historical contacts recovered into attribution and made visible in all reporting.
- Idempotent MQL classification → routing → rep notification pipeline live end-to-end.

STACK

HubSpot chatbot + workflows

n8n

Azure OpenAI

HubSpot CRM API

Slack notifications

WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
Attribution flag captured chat data but never stamped the source	Built a stamp workflow for go-forward + API backfill for history. Prevention: audit the whole capture→report chain, not just capture.

05

DEPLOY-READY

Partner PQL Workflow

Classifies inbound partner-form submissions, drafts a partner-specific email, and routes a task to the partnerships owner automatically.

THE PROBLEM

Partner inquiries needed manual triage, classification, and routing before anyone could act.

WHAT I BUILT

- An n8n workflow: form webhook → AI classify (partner vs. not) → AI-drafted partner email → lifecycle set → CRM task routed to the partnerships owner.
- Hardened for production: fixed a v3-API association bug (id vs. vid) that would have created tasks with no contact link, and wired a shared error-handler workflow.

SYSTEMS THINKING

Explicitly matched the error-handling pattern of the existing inbound workflow so failures surface in one place — thinking about the operational estate, not just the single workflow.

OUTCOME

- JSON is deploy-ready; owner ID patched, association bug fixed, error workflow wired. Blocked only on four CRM properties + a partner-form change on another owner's track.

STACK

n8n

Azure OpenAI (gpt-5-chat)

HubSpot API

WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
HubSpot v2.2 nodes use the v3 API, which returns id not vid	Read id first with a vid fallback, so tasks always associate to the contact. Prevention: verify API-version field shapes on import.

Entity Readiness Change Report

Every Monday, diffs the company's country hire-readiness against last week and emails Growth only the countries that flipped — as a locked two-column table.

THE PROBLEM

Nobody caught weekly changes in which countries the company could legally hire in. Marketing pays for country-specific search ads and didn't want to keep bidding on a country it could no longer serve — or miss one that just opened.

WHAT I BUILT

- An importable n8n workflow: weekly trigger → pull readiness → normalize to a boolean → diff vs. saved baseline → report only the flips → deliver → then write the new baseline.
- A hard no-cloud constraint honored: state lives in an n8n Data Table (source of truth) mirrored to an Excel file for humans — no Azure.

SYSTEMS THINKING

Two constraints drove the architecture and were respected exactly: baseline writes happen only after a successful, non-empty pull (a failed run never corrupts last week's state), and the output stays locked to two columns per the stakeholder's requirement. First-run seeding, empty-pull, and failed-pull edge cases are each handled.

OUTCOME

- Build-ready spec + importable workflow delivered; routed to VP Growth + reviewer for sign-off per the approval gate. Phase-2 agency feed built but held pending format spec.

STACK

n8n

Onspring

n8n Data Table

Microsoft Excel / OneDrive

Outlook / Teams

WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM

Risk: a failed or empty data pull could overwrite good state

ROOT-CAUSE FIX & PREVENTION

Baseline writes gated behind a successful non-empty pull; normalize step throws on empty. Prevention: never let a failed run mutate the source of truth.

SDR-Agency Prospecting Enablement (AGS)

Runs the full prospecting supply chain for an outsourced SDR agency — clean, enriched, CRM-safe lists activated on a weekly cadence.

THE PROBLEM

A paid sales agency needed a steady stream of clean, targeted, compliance-screened lists — without contacting existing customers, open deals, closed-lost, or recently-touched contacts.

WHAT I BUILT

- A repeatable intake → normalize → dedupe → enrich → suppress → activate pipeline across Clay, Amplemarket, ZoomInfo and HubSpot.
- Live HubSpot suppression screening for customer status, open deals, and closed-lost on every company before send.
- A competitive-takeaway play: 188 verified HR contacts at companies currently on a named competitor EOR, ~100 net-new to CRM.

SYSTEMS THINKING

Treats prospecting as a governed data pipeline with source-of-truth boundaries (CRM = truth, enrichment = workspace, orchestration = glue) and a compliance-first suppression layer — the difference between a list and a liability.

OUTCOME

- 5,398-contact master activated; weekly batches tagged by segment for agency-level reporting.
- Every list screened live against the CRM — 4 companies pulled for open-deal/closed-lost, 20 contacts removed for active-sequence/recent-touch before send.

STACK

Clay

Amplemarket

ZoomInfo

HubSpot API

WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
Parallel one-off workbooks caused drift and duplicated effort	Consolidated into one master workbook + segment tabs as the operating source of truth.

Outbound Signals & Intent Engine

A defined library of fit + intent signals, a 0–100 account scoring model, and composite “plays” that stack signals into prioritized, repeatable outbound motions.

THE PROBLEM

Broad, untargeted outbound was expensive and easy to ignore, with no defensible reason for any given touch.

WHAT I BUILT

- Eight signal categories (expansion, funding, hiring, leadership, competitor/tech, compliance, first-party engagement, suppression), each weighted and mapped to a data source, persona, and query logic.
- A flagship “Displacement” play targeting Director+ People leaders who hire internationally and already use a competitor EOR.
- A companion Master Signal & Intent Library workbook feeding Clay build sessions.

SYSTEMS THINKING

Turns intuition into a system: every touch has a named, weighted, time-bound reason behind it, which is what lifts reply rates and keeps enrichment credit and rep time focused on accounts with a real reason to buy now.

OUTCOME

- Framework + scoring model + flagship play documented for GTM leadership and operationalized in Clay.

STACK

Clay

Factors.ai

HubSpot

Scoring model (0–100)

Clay Credit Governance & Cost Control

Traced a runaway enrichment-credit spike to its real sources and shipped a reusable template + guardrails that cut per-row cost and stop the leak repeating.

THE PROBLEM

Enrichment credit usage spiked from <15K/month to 515K in a single month. 21% of a month's spend ran through unattributed "Copy of" tables, phone numbers were bought 2–3x per contact, and people-searches ran with no ICP gate.

WHAT I BUILT

- A full-year forensic audit (a seven-sheet master dataset) that re-attributed the unnamed "Copy of" spend back to its real source tables.
- A Standard Enrichment Table Template: a free ICP-fit gate in front of every paid column, a 3-column deduped phone model capped at 3 attempts, and AI/HTTP columns gated to top-tier accounts.
- Three build profiles (Lean ~15 / Balanced ~21 / Full ~50 credits per row) replacing an undisciplined ~50-credit average.

SYSTEMS THINKING

A credit-planning architecture, not a one-time cleanup: gate → cap → clone-don't-copy → dedupe → consolidate → review weekly. The weekly review replaces after-the-fact audits so the next spike is caught in days, not months.

OUTCOME

- Root cause identified and a leadership one-sheeter + master dataset delivered to VP Growth, replacing a pile of ad-hoc exports.
- Per-row cost brought under explicit control; a standing weekly credit review installed as the go-forward guardrail.

STACK

Clay API

Excel data modeling

PDF one-sheeter for leadership

WHAT BROKE & HOW IT WAS FIXED

ISSUE / SYMPTOM	ROOT-CAUSE FIX & PREVENTION
A single day burned 201,924 credits from a phone-waterfall re-buying the same numbers	Split phone into three deduped columns capped at 3 provider attempts. Prevention: dedupe + cap before any waterfall runs.

10

META-SYSTEM · FRAMEWORK

New Project Automation System (Center of Excellence)

Turns any raw input — notes, files, links, a one-line idea — into a complete, leadership-ready project package: knowledge base, PRD, architecture diagram, reporting framework, and a phased build plan.

THE PROBLEM

Projects were scattered across a task tool (inconsistent), chat threads (buried), and nowhere (lost). Leadership had no clean answer to “what are we building, why, and when.” Every project was an island only one person understood.

WHAT I BUILT

- An intake engine built on a documented framework (PAL for depth, NPAO for work classification, 4Ds for phasing, RAG/DAL for provenance) that outputs a standardized package every time.
- A searchable project index/database so any initiative is one link away for leadership and onboarding.
- The engine that spawned the reusable skills below — projects come out clean, complete, and consistent.

SYSTEMS THINKING

This is the systems-thinking artifact of the whole portfolio: a repeatable operating model that standardizes how an entire team scopes, documents, and ships work — designed so quality is structural, not dependent on any one person.

OUTCOME

- Idea → fully-scoped project cut from 4–8 hours to under 30 minutes.
- A permanent, searchable Center of Excellence: consistent PRDs, architecture, and build plans across every GTM initiative.

STACK

Claude / Agent SDK

n8n

Notion

Asana

RAG web search

GTM Skill & Agent Library (30+)

A library of reusable AI skills and agents that put every automation above into the hands of non-technical reps and marketers through plain-English commands.

THE PROBLEM

Powerful automations are useless if only the builder can run them. A 50+ person GTM org needed self-serve, rep-friendly tools — not one-off scripts.

WHAT I BUILT

- Prospecting & enrichment: Clay prospecting engine, credit estimator, CSV enrichers, HubSpot import formatter, list routers.
- Workflow & ops: n8n workflow architect, execution analyst, GTM architect, project intake & handoff, close-out reporting.
- Revenue & enablement: HubSpot data/revenue analysts, RFP knowledge-base builder + answer retriever, Factors.ai signal operator, case-study builder.
- Content at scale: a full branded content suite (email, social, ads, blog, decks, one-pagers) plus an agent-factory that builds new skills from a description.

OUTCOME

- 30+ skills/agents shipped and organized into installable suites for the GTM team.
- Repeatable, governed self-service replacing builder-dependent one-offs.

STACK

Claude Skills / Agent SDK

n8n

Clay

HubSpot

Factors.ai

Avoma

SYSTEMS THINKING

A platform play: reusable skills with clear triggers, composability (skills call specialist skills), and guardrails baked in (secret-handling, CRM-safety, approval gates) — so scaling to 50+ users doesn't scale the risk.

Patrick Diamitani

GTM AI & Automation Manager · Atlas HXM · Chicago, IL

pdiamitani@atlashxm.com · linkedin.com/in/diamitani

Every metric and technical detail on this page is drawn from real, shipped project records. Full edition — shared under NDA / for direct evaluation. No secrets, keys, tokens, or endpoints are reproduced anywhere in this document.